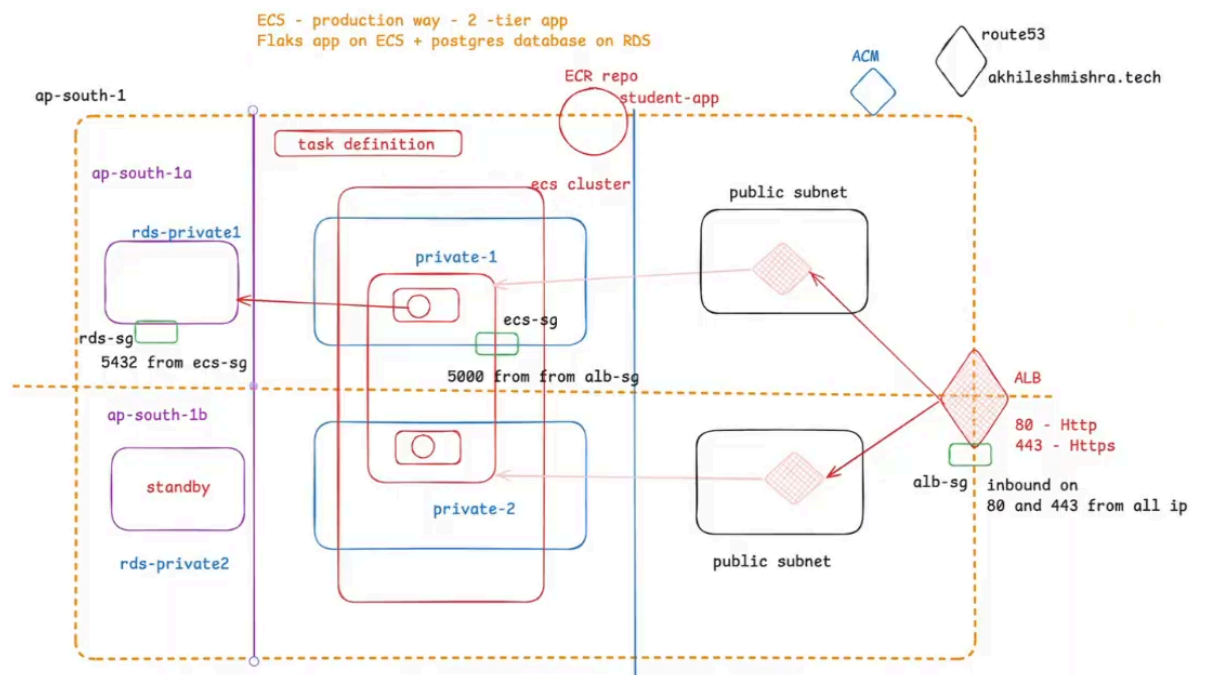
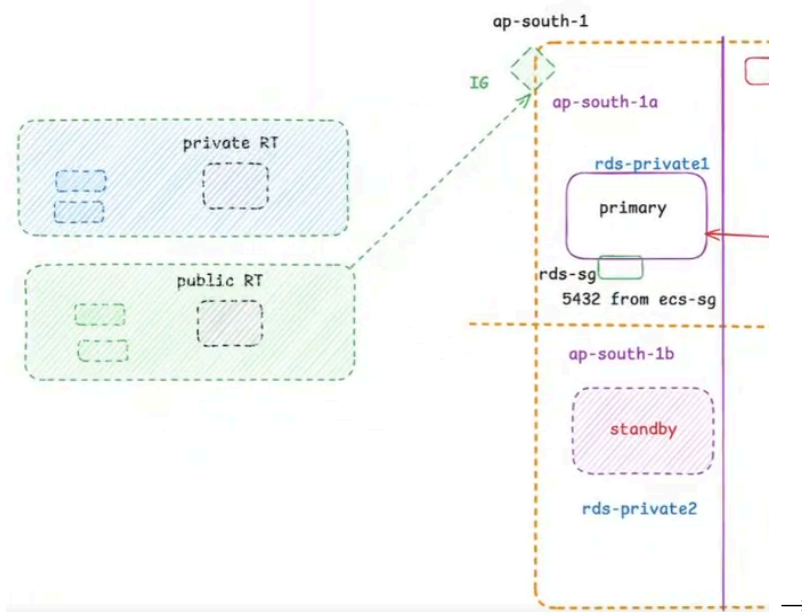


Production Ready ECS



VPC

First is necessary to create a VPC to hold our infrastructure, it will be composed of:

- 1 VPC
- 2 Private subnets
- 2 Subnets for database
- 2 Public subnets for receiving traffic and ALB

```

# vpc

dev-vpc - 10.0.0.0/16

## subnets

private1 - 10.0.1.0/24
private2 - 10.0.2.0/24

public1 - 10.0.3.0/24
public2 - 10.0.4.0/24

# rds subnts
rds1 - 10.0.5.0/24

rds 2 - 10.0.6.0/24

```

It can be done at “VPC” section and subnets sections, here will always use the same two availability zones for practice purposes (Is necessary to select at least two availability zones for the subnets in order to be able to create the RDS databases with high availability).

Associated VPC CIDRs

IPv4 CIDRs
10.0.0.0/16

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 6

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
private1
The name can be up to 255 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
United States (N. Virginia) / us-east-1a

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must be within this block.
10.0.0.0/16

IPv4 subnet CIDR block
10.0.1.0/24 255 IPs

RDS

Now will start creating the database, it's not strictly necessary to proceed in this order to create the components of our app but will try to follow a sequence:

1. VPC and subnets
2. DB subnet group
3. RDS
4. Route tables

5. Internet Gateway
6. Security Groups
7. Secrets Manager
8. ECR
9. Docker Image
10. IAM roles
11. ECS cluster
12. Task definition
13. NAT Gateway
14. Route table route to NAT
15. ALB
16. Target Group
17. ECS Service
18. Domain (ACM, Route 53)
19. Logs

First will need to create the subnet groups necessary to create the rds after with high availability, remember the vps has to have multiple availability zones within a region (subnets in multiple az's).

Choose a VPC Identifier that corresponds to the subnets you want to use for your DB subnet group. You won't be able to choose a different VPC identifier after your subnet group has been created.

week5-vpc (vpc-0ca1e82905ce0556c)
6 Subnets, 2 Availability Zones

Add subnets

Availability Zones
Choose the Availability Zones that include the subnets you want to add.

Choose an availability zone

us-east-1a ✕ us-east-1b ✕

Subnets
Choose the subnets that you want to add. The list includes the subnets in the selected Availability Zones.

Select subnets

rds1
Subnet ID: subnet-0578ff49f8303ff25 CIDR: 10.0.5.0/24 ✕

rds2
Subnet ID: subnet-0a2e161f69746e2e5 CIDR: 10.0.6.0/24 ✕

ⓘ For Multi-AZ DB clusters, you must select 3 subnets in 3 different Availability Zones.

Subnets selected (2)

Availability zone	Subnet name	Subnet ID	CIDR block
us-east-1a	rds1	subnet-0578ff49f8303ff25	10.0.5.0/24
us-east-1b	rds2	subnet-0a2e161f69746e2e5	10.0.6.0/24

Cancel Create

Now, go to “Aurora and RDS” > Subnet Groups > Create DB subnet group

In order to create the rds database go to “Aurora and RDS” section, then create database.

Engine options

Engine type [Info](#)

☐ Aurora (MySQL Compatible)

☐ Aurora (PostgreSQL Compatible)

☐ MySQL

☒ PostgreSQL

☐ MariaDB

☐ Oracle

☐ Microsoft SQL Server

☐ IBM Db2

Select postgres, not the latest engine to avoid unsolved version bugs.

Contemplate will use free tier to avoid charges.

In the settings will be using a self managed password for now, later will use AWS Secrets manager.

The screenshot shows the 'Settings' page for an AWS RDS database instance. The 'DB instance identifier' is set to 'database-1'. Under 'Credentials Settings', the 'Master username' is 'postgres'. In the 'Credentials management' section, the 'Self managed' option is selected. The 'Master password' is masked with dots, and the 'Password strength' indicator shows 'Very weak'. The 'Confirm master password' field is also masked.

Settings

DB instance identifier [Info](#)
Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.
database-1
The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

Credentials Settings

Master username [Info](#)
Type a login ID for the master user of your DB instance.
postgres
1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management
You can use AWS Secrets Manager or manage your master user credentials.

☐ Managed in AWS Secrets Manager - most secure
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ Self managed
Create your own password or have RDS create a password that you manage.

☐ Auto generate password
Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)
[Masked password]
Password strength **Very weak**
Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / * @

Confirm master password [Info](#)
[Masked password]

Identifier database. Username postgres. Password is 12345678 for practice. 5432 database port. studentportal database name.

The screenshot shows the 'Storage' configuration page for an AWS RDS database instance. The 'Instance type' is 'db.t3.micro'. The 'Storage type' is 'General Purpose SSD (gp2)'. The 'Allocated storage' is '20 GiB'. The 'Maximum storage threshold' is '1000 GiB'. The 'Enable storage autoscaling' checkbox is checked.

Instance type
db.t3.micro
2 vCPUs 1 GiB RAM EBS Bandwidth: Up to 2085 Mbps Network: Up to 5 Gbps

Storage

Storage type [Info](#)
Provisioned IOPS SSD (io2) storage volumes are now available.
General Purpose SSD (gp2)
Baseline performance determined by volume size

Allocated storage [Info](#)
20 GiB
Allocated storage value must be 20 GiB to 6144 GiB

Additional storage configuration

Storage autoscaling [Info](#)
Provides dynamic scaling support for your database's storage based on your application's needs.
☒ Enable storage autoscaling
Enabling this feature will allow the storage to increase after the specified threshold is exceeded.

Maximum storage threshold [Info](#)
Charges will apply when your database autoscales to the specified threshold
1000 GiB
Allocated storage value must be 22 GiB to 6144 GiB

Connectivity Info

Compute resource
Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☒ **Don't connect to an EC2 compute resource**
 Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☐ **Connect to an EC2 compute resource**
 Set up a connection to an EC2 compute resource for this database.

Virtual private cloud (VPC) Info
Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

Default VPC (vpc-0e817f0e77d72b1e4) 0 Subnets, 0 Availability Zones

After a database is created, you can't change its VPC. Only VPCs with a corresponding DB subnet group are listed.

ⓘ After a database is created, you can't change its VPC.

DB subnet group Info
Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

default

Public access Info

☐ **Yes**
 RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.

☒ **No**
 RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall) Info
Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☒ **Choose existing**
 Choose existing VPC security groups

☐ **Create new**
 Create new VPC security group

Later will create the security groups.

In the availability zone select the main availability zone (example: us-east-1a).

Route tables

Let's create private and public route tables in order to set rules and indicate where the traffic has to go, in this case we want that public subnets connect to the internet (gateway) and private ones to indicate ECS that can connect to the NAT gateway (later).

rds1 and rds2 do not need internet routes. They still belong to the VPC routing domain and use the local route automatically, but they do not require a public route to an Internet Gateway or a default route to a NAT Gateway for this exercise, so let's associate like this:

rtb-0ca9ef6e9577b3552 / week5-privatert Actions

Details Info Route table ID rtb-0ca9ef6e9577b3552 VPC vpc-0ca1e82905ce0556c week5-vpc		Main No Owner ID 166337233399	Explicit subnet associations 2 subnets	Edge associations -
---	--	--	---	------------------------

Routes | **Subnet associations** | Edge associations | Route propagation | Tags

Explicit subnet associations (2) Edit subnet associations

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
private2	subnet-05c1c06af46a2feb5	10.0.2.0/24	-
private1	subnet-0c7ff4af59371a937	10.0.1.0/24	-

Subnets without explicit associations (4) Edit subnet associations

The following subnets have not been explicitly associated with any route tables and are therefore associated with the main route table:

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
public2	subnet-0429b154062ab6fdb	10.0.4.0/24	-
rds1	subnet-0578ff49f8303ff25	10.0.5.0/24	-
rds2	subnet-0a2e161f69746e2e5	10.0.6.0/24	-
public1	subnet-07f5f580df4a192fc	10.0.3.0/24	-

Now let's create the public one, that will be associated to the public subnets and attached to the Internet gateway in the next section:

You have successfully updated subnet associations for rtb-01d512c7e97a8dfa8 / week5-publicrt.

rtb-01d512c7e97a8dfa8 / week5-publicrt

Actions

Details

Route table ID
rtb-01d512c7e97a8dfa8

VPC
vpc-0ca1e82905ce0556c / week5-vpc

Main

No

Owner ID
166337233399

Explicit subnet associations

2 subnets

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Explicit subnet associations (2)

Edit subnet associations

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
public2	subnet-0429b154062ab6fdb	10.0.4.0/24	-
public1	subnet-07f5f580df4a192fc	10.0.3.0/24	-

Subnets without explicit associations (2)

Edit subnet associations

The following subnets have not been explicitly associated with any route tables and are therefore associated with the main route table.

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
rds1	subnet-0578ff49f8303ff25	10.0.5.0/24	-
rds2	subnet-0a2e161f69746e2e5	10.0.6.0/24	-

Internet Gateway

We already have our subnets created and the route tables, now we want resources in the public subnets, such as the Application Load Balancer, to be reachable from the internet, therefore we will create the internet gateway and later attach the public route table to the gateway:

Create internet gateway

An Internet gateway is a virtual router that connects a VPC to the Internet. To create a new Internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag
Creates a tag with a key of 'Name' and a value that you specify.
week5-igw

Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
Key: Name, Value: week5-igw
Add new tag
You can add 49 more tags.

Cancel

Create internet gateway

Attach to VPC (igw-05890fff94bfc035)

VPC
Attach an Internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.
Available VPCs
Attach the Internet gateway to this VPC.
vpc-0ca1e82905ce0556c
AWS Command Line Interface command

Cancel

Attach internet gateway

After attach we need to tell the subnet that this is the “route” to get connections from internet, so we edit our public route table to have access to the internet gateway:

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	-	No	CreateRoute

Buttons: Add route, Cancel, Preview, Save changes, Remove

We can test connection creating a EC2 in the public subnet and connecting, but let's continue in order to not consume too much time.

Security groups

Now, we want certain connections to be allowed inside our VPC and also from outside, this connections are:

- ALB has to connect to the IGW that allows connections from outside
- RDS has to connect to the ECS cluster
- ECS needs to connect to the ALB (receive connections)

So, this would be the rules in the security groups:

ALB:

- Inbound: HTTP 80:0.0.0.0/0
- Inbound: HTTPS 443:0.0.0.0/0
- Outbound: All allowed

ECS:

- Inbound: 5000(app port):*ALBSECURITYGROUP*
- Outbound: All allowed

RDS:

- Inbound: 5432(Postgres):*ECSSECURITYGROUP*
- Outbound: Default

As we already have the rds created we can add the security group to it:

Connectivity

Network type Info
To use dual-stack mode, make sure that you associate an IPv6 CIDR block with a subnet in the VPC you specify.

☒ IPv4
Your resources can communicate only over the IPv4 addressing protocol.

☐ Dual-stack mode
Your resources can communicate over IPv4, IPv6, or both.

DB subnet group
week5

Security group
List of DB security groups to associate with this DB instance.
Choose security groups
rds-securitygroup X

Certificate authority Info
Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision.
rds-ca-rsa2048-g1 (default)
Expiry: May 26, 2061

Secrets Manager

Before my setup was a self managed key, but now lets use secrets manager that is more robust in AWS:

The screenshot shows the AWS Secrets Manager console during the 'Choose secret type' step. On the left, a progress bar indicates the steps: Step 1 (Choose secret type), Step 2 (Configure secret), Step 3 (optional: Configure rotation), Step 4 (Review). The main area has three sections: 'Secret type' with radio buttons for 'Credentials for Amazon RDS database', 'Credentials for Amazon DocumentDB database', 'Credentials for Amazon Redshift data warehouse', 'Credentials for other database', 'Managed external secret', and 'Other type of secret' (selected); 'Key/value pairs' with a table showing a key 'DBLINK' and a value 'postgresql://postgres:12345678@database-w5.cocb7dorigmz.us-east-1.rds.amazonaws.com'; and 'Encryption key' with a dropdown set to 'aws/secretsmanager' and a 'Add new key' link. 'Cancel' and 'Next' buttons are at the bottom right.

Here we use full link to match with the app config:

The screenshot shows a code editor with a file explorer on the left and a Python file named 'config.py' open. The file explorer shows a project structure with folders 'week1' through 'week4' and a 'week5 / student-portal' folder containing subfolders 'app', 'models', 'routes', 'static', and 'templates', along with files '__init__.py', 'logging_config.py', 'metrics.py', 'config.py', 'Docker-compose.yml', 'Dockerfile', 'readme.md', 'requirements.txt', 'run.py', and 'README.md'. The 'config.py' file is selected and its content is displayed in the editor. The code includes an import for 'os', a variable assignment, and a 'Config' class with attributes for 'SQLALCHEMY_DATABASE_URI' and 'SQLALCHEMY_TRACK_MODIFICATIONS'. The terminal at the bottom shows the command prompt 'miguel@miguel-HP-ENVY-TS-17-Notebook-PC:~/Documentos/DevopsBootcamp/devops_projects\$'.

```
1 import os
2
3 variable_which_i_wont_use = "this is a variable which i wont use"
4
5
6 class Config:
7     SQLALCHEMY_DATABASE_URI = os.getenv("DB_LINK")
8     SQLALCHEMY_TRACK_MODIFICATIONS = False
9     Ctrl+L to chat, Ctrl+K to generate
```

Which is called by the init of the app (__init__.py). Now go to the RDS and setup the secret:

Credentials management
You can use AWS Secrets Manager or manage your master user credentials.

☒ **Managed in AWS Secrets Manager - most secure**
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☐ **Self managed**
Create your own password or have RDS create a password that you manage.

ⓘ If you manage the master user credentials in AWS Secrets Manager, additional charges apply. See [AWS Secrets Manager pricing](#). Additionally, some RDS features aren't supported. See [limitations here](#).

Select the encryption key | Info
You can encrypt using the KMS key that Secrets Manager creates or a customer managed KMS key that you create.

aws/secretsmanager (default)

[Add new key](#)

▼ **Additional credentials settings**

Database authentication options | Info

☒ **Password authentication**
Authenticates using database passwords.

☐ **Password and IAM database authentication**
Authenticates using the database password and user credentials through AWS IAM users and roles.

☐ **Password and Kerberos authentication**
Choose a directory in which you want to allow authorized users to authenticate with this DB instance using Kerberos Authentication.

ECR

Now let's create the ECR for register the Docker image:

Amazon ECR > Private registry > Repositories > Create private repository

Create private repository

General settings

Repository name
Enter a concise name. Repositories support namespaces, which you can use to group similar repositories.
166337233399.dkr.ecr.us-east-1.amazonaws.com/week5-repository
16 out of 256 characters maximum (2 minimum). The name must start with a letter and can only contain lowercase letters, numbers, and special characters (.,/).

Image tag settings | Info

Image tag mutability
Choose the tag mutability setting.
☒ **Mutable**
Image tags can be overwritten.
☐ **Immutable**
Image tags can't be overwritten.

Mutable tag exclusions
Tags that match these filters will be immutable (can't be overwritten). Using wildcards (*) will match zero or more image tag characters.

Filters must only contain letters, numbers, and special characters (.,/). Each filter is limited to 128 characters, 2 wildcards (*), and you can add up to 5 filters in the exclusion list. [Add filter](#)

Docker Image

Having created the repository, let's now build and register the docker image:

First to login (will ask to open a tab in chrome to retrieve credentials):

aws ecr get-login-password --region us-east-1 | docker login --username AWS --password-stdin
166337233399.dkr.ecr.us-east-1.amazonaws.com

Then build the image:

docker build -t week5-repository .

Tag the image:

docker tag week5-repository:latest
166337233399.dkr.ecr.us-east-1.amazonaws.com/week5-repository:latest

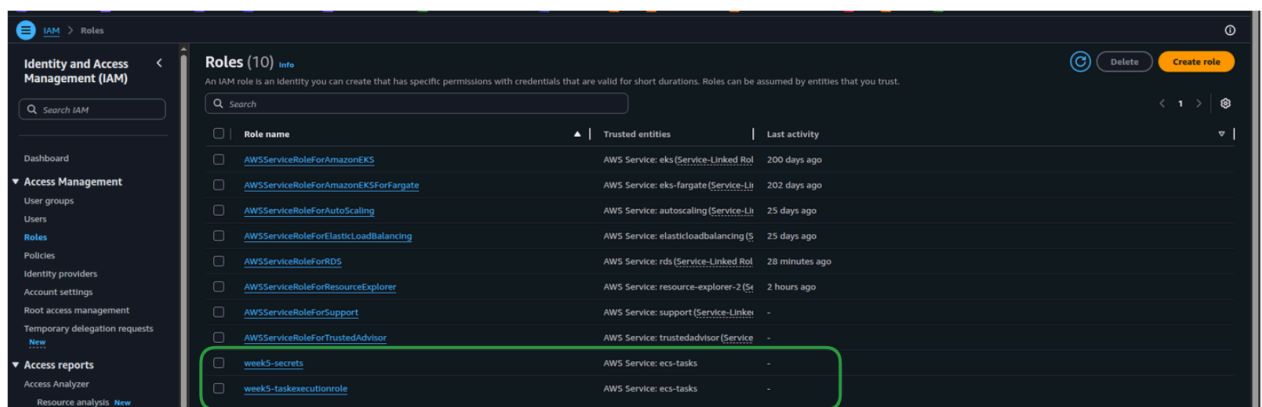
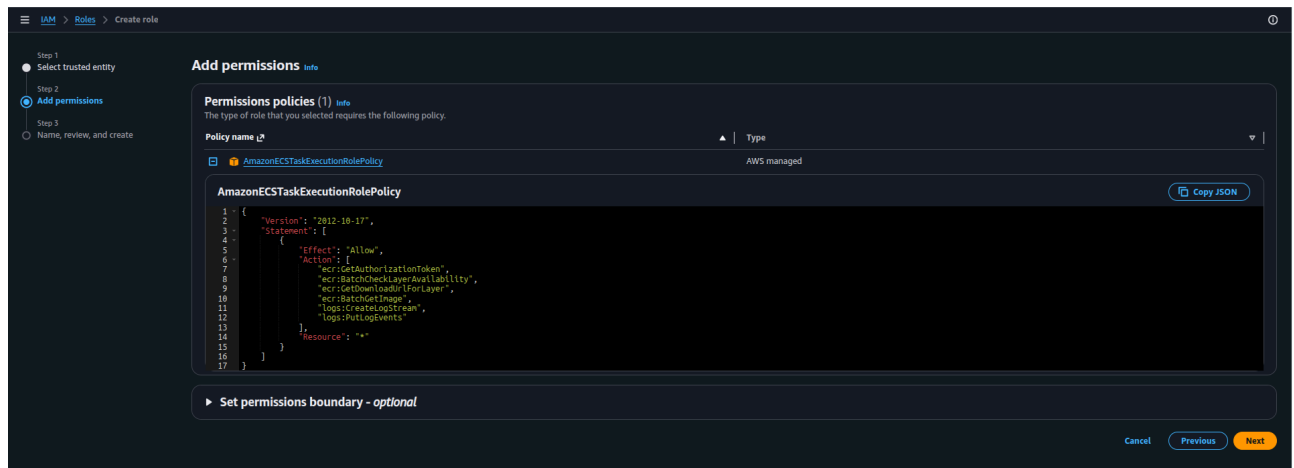
Push the image:

docker push 166337233399.dkr.ecr.us-east-1.amazonaws.com/week5-repository:latest

IAM roles

Now we need two IAM roles, access to read secrets from the app (ECS) and task execution role, lets create them:

1. Trusted Entity: AWS Service > Use Case: Elastic Container Service – Task Execution
2. Trusted Entity: AWS Service > Use Case: Elastic Container Service – Task Execution



ECS cluster

Now lets create the ECS cluster, the service which runs the application. Remember there are two options, using EC2 or Fargate (serverless), each of them hve pros and cons, in this case will use Fargate because of its simplicity to set up.

Create cluster > Fargate Only > Create service (Important park comes in the task definition)

Task Definition

Now lets create the task definition, remember to use the roles, task execution was created before and allows ecs to execute the app, task role allows containers inside the task to do things, in this case to use secrets manager to login to the database.

Amazon Elastic Container Service > Create new task definition

Amazon Elastic Container Service

Express Mode
Clusters
Namespaces
Task definitions
Account settings

Amazon ECR
Repositories
AWS Batch
Online learning workshop
Documentation
Discover products
Subscriptions

☐ **Managed instances**
Use if you have specific hardware constraints, such as GPU accelerators, CPU instruction sets, or network-optimized hardware (offloads scaling, patching, and instance management to AWS).

☐ **Amazon EC2 instances**
Self-managed infrastructure using Amazon EC2 instances.

OS, Architecture, Network mode
Network mode is used for tasks and is dependent on the compute type selected.

Operating system/Architecture **Info**
Linux/X86_64

Network mode **Info**
awsipc

Task size **Info**
Specify the amount of CPU and memory to reserve for your task.

CPU
5 vCPU

Memory
1 GB

Task roles - conditional

Task role **Info**
A task IAM role allows containers in the task to make API requests to AWS services.

week5-secrets  [Create new role](#)

Task execution role **Info**
A task execution IAM role is used by the container agent to make AWS API requests on your behalf. If you don't already have a task execution IAM role created, we can create one for you.

week5-taskexecutionrole  [Create new role](#)

The port used is the port where the app is deployed:

Port mappings **Info**
Add port mappings to allow the container to access ports on the host to send or receive traffic. For port name, a default will be assigned if left blank.

Container port	Protocol	Port name	App protocol	
5000	TCP	app-port	HTTP	Remove

[Add port mapping](#)

In the environment variables use the arn from the secret created before:

AWS Secrets Manager > Secrets > rds-w5-secret

rds-w5-secret

Secret details  [Actions](#)

Encryption key
aws/secretsmanager

Secret name
rds-w5-secret

Secret ARN
arn:aws:secretsmanager:us-east-1:166337233399:secret:rds-w5-secret-qgNVp

Secret description
-

Secret type
-

Amazon Elastic Container Service > Create new task definition

Amazon Elastic Container Service

Express Mode
Clusters
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Account settings

Environment variables - optional

Environment variables **Info**
Add individually
Add environment variables using plain text values or secrets from AWS Secrets Manager or Parameter Store.

Key	Value type	Value	
DB_LINK	ValueFrom	arn:aws:secretsmanager:us-east-1	Remove

[Add environment variable](#)

Then review task definition and run to see what breaks:

▼ Networking

VPC [Info](#)
Select a VPC to use for your Amazon ECS resources.

vpc-0ca1e82905ce0556c
week5-vpc

[Create a new VPC](#)

Subnets
Choose the subnets within the VPC that the task scheduler should consider for placement.

Choose subnets

Clear current selection

subnet-05c1c06af46a2feb5
private2
us-east-1b 10.0.2.0/24

subnet-0c7ff4af59371a937
private1
us-east-1a 10.0.1.0/24

Security group [Info](#)
Choose an existing security group or create a new security group.

☒ Use an existing security group
☐ Create a new security group

Security group name
Choose an existing security group.

Choose security groups

sg-02fc519d0aa350f30
ecs-securitygroup

Public IP [Info](#)
Choose whether to auto-assign a public IP to the task's elastic network interface (ENI).

☐ Turned off

If you get this error:

Amazon Elastic Container Service > Clusters > week5-ecs > Tasks > b06db15f836340c682aef3c589bdb933 > Configuration

Tasks launched

am:aws:ecs-us-east-1:166337233399:task/week5-ecs/b06db15f836340c682aef3c589bdb933

b06db15f836340c682aef3c589bdb933

Task stopped at: 2026-03-15T09:26:48.812Z

ResourceInitializationError: unable to pull secrets or registry auth: unable to retrieve secret from arn: There is a connection issue between the task and AWS Secrets Manager. Check your task network configuration. failed to fetch secret arn:aws:secretsmanager:us-east-1:166337233399:secret:ids=w5-secret-gpiVip from secrets manager: operation error Secrets Manager: GetSecretValue, https response error StatusCode: 0, RequestID: , canceled, context deadline exceeded

[View troubleshooting guide](#)

[Inspect with Amazon Q](#)

Configuration Metrics Logs Networking Volumes (0) Tags

Container details for appstudent

It's because of the NAT Gateway, we don't have connection to the internet as it is a private network so we want it now in order to be able to pull the image from the ECR registry.

NAT Gateway

NAT gateway settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.
natgw-week5
The name can be up to 256 characters long.

Availability mode Info
Choose whether to deploy across all zones in the region or restrict to a single availability zone.

☐ Regional - new
Scales automatically across all regional AZs, simplifying management for multi AZ deployments.

☒ Zonal
Provides granular control within a specific availability zone, adhering to subnet level settings.

Subnet
Select a subnet in which to create the NAT gateway.
subnet-07f5f80df4a192fc (public1)

Connectivity type
Select a connectivity type for the NAT gateway.
☒ Public
☐ Private

Elastic IP allocation ID Info
Assign an Elastic IP address to the NAT gateway.
eipalloc-037f62b23905f95c Allocate Elastic IP

[Additional settings](#) Info

Remember NAT Gateway has to be on the public subnet.

The add the NAT to the private route table:

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	NAT Gateway	-	No	CreateRoute

Add route Cancel Preview Save changes

As it's connected to the private subnet already, it will allow the download of the image and secrets.

If you get this error:

Task stopped at 2026-03-15T09:46:33.595Z Inspect with Amazon

ResourceInitializationError: unable to pull secrets or registry auth: execution resource retrieval failed: unable to retrieve secret from aws: service call has been retried 1 time(s): failed to fetch secret arn:aws:secretsmanager:us-east-1:16637233399:secret:ids-w5-secret-qgiVp from secrets manager: operation error Secrets Manager: GetSecretValue, https response error StatusCode: 400, RequestID: b640b37c-96ec-4485-a0b6-35d8452e2739, api error AccessDeniedException: User: arn:aws:sts::16637233399:assumed-role/week5-taskexecutionrole/59dc95e8171e4c279a325e87c9e94b3a is not authorized to perform: secretsmanager:GetSecretValue on resource: arn:aws:secretsmanager:us-east-1:16637233399:secret:ids-w5-secret-qgiVp because no identity-based policy allows the secretsmanager:GetSecretValue action

[View troubleshooting guide](#)

Add the policy to read secrets to the role:

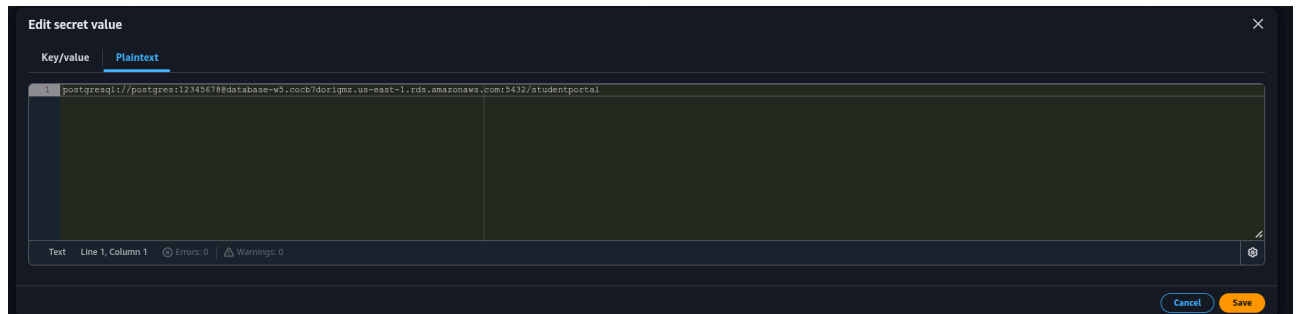
Permissions policies (2) Info Simulate Remove Add permissions

You can attach up to 10 managed policies.

Filter by Type: All types

Policy name	Type	Attached entities
AmazonECSTaskExecutionRolePolicy	AWS managed	1
AWS Secrets Manager Client Read Only Access	AWS managed	2

As the app is returning an error parsing the key/value secret, will use plaintext in the secret for practice purposes



In this case as I used the managed password in AWS Secrets, I had to change the password and add ssl for another error I had:

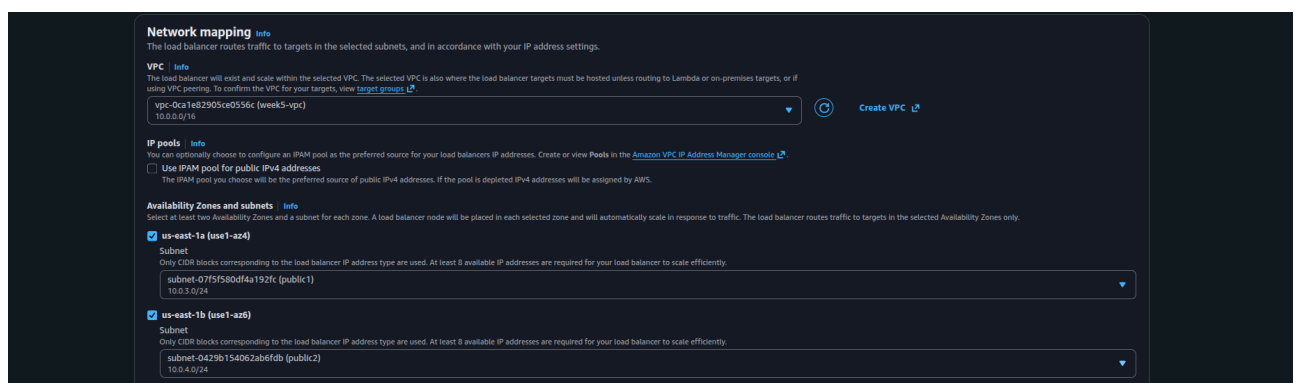
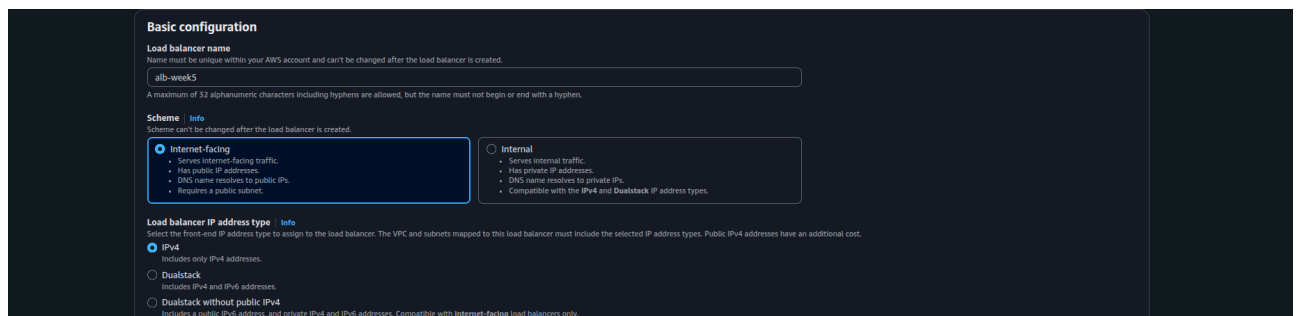
postgresql://postgres:oHM%28p8X%3Ek%7C_%29hvAcjS4Pr%24jPRb%29O@database-w5.cocb7dorigmz.us-east-1.rds.amazonaws.com:5432/studentportal?sslmode=require

This was added to the secret I created as plain text.

Now I have “running” in the task, lets now be able to connect through ALB.

ALB & Target Group

Lets create the ALB:



Step 1: Create target group

Step 2: recommended

Step 3: Register targets

Step 4: Review and create

Review and create

Review your target group configuration before creating

Step 1: Target group details

Target group details			
Name week5-tg	Target type IP	Protocol : Port HTTP: 8000	Protocol version HTTP1
VPC vpc-dca1ed22055cd0556c	IP address type IPv4		

Health check details

Health check details			
Health check protocol HTTP	Health check path /	Health check port traffic-port	Interval 30 seconds
Timeout 5 seconds	Healthy threshold 5	Unhealthy threshold 2	Success codes 200

Step 2: Register targets

Targets (0)		
IP address	Port	Zone
No targets added		

Cancel Previous **Create target group**

Now continue with ALB:

Create Certificate Manager > Register in Route 53 (Done in previous labs)

EC2 > Load balancers > alb-week5

Introducing ALB target optimizer
Target optimizer lets you enforce a maximum number of requests per target using an ALB-provided agent, improving success rates, latency, and efficiency. [Learn more](#)

alb-week5

Details

Load balancer type	Status	VPC	Load balancer IP address type
Application	Provisioning	vpc-dca1ed22055cd0556c	IPv4
Scheme	Hosted zone	Availability Zones subnet-0429b154062ab6fdb us-east-1b (use1-azb) subnet-07f5f980d4e192fc us-east-1a (use1-aza)	Date created March 15, 2026, 12:30 (UTC+01:00)
Internet-facing	Z35XMDTRQ7X7K		

Load balancer ARN
[arn:aws:elasticloadbalancing:us-east-1:166337233339:loadbalancer/app/alb-week5/e6e679d68aa317a0](#)

DNS name info
[alb-week5-352477752.us-east-1.elb.amazonaws.com](#) (A Record)

Listeners and rules (2)

A listener checks for connection requests on its configured protocol and port. Traffic received by the listener is routed according to the default action and any additional rules.

Protocol/Port	Default action	Rules	ARN	Security policy	Default SSL/TLS certificate	mTLS	Trust store
HTTP:443	Forward to target group week5-tg 1 (100%) Target group stickiness: Off	1 rule	ARN	ELBSecurityPolicy-TLS13-1-2-...	altizovh (Certificate ID: fcd459...	Off	Not applicable
HTTP:80	Forward to target group week5-tg 1 (100%) Target group stickiness: Off	1 rule	ARN	Not applicable	Not applicable	Not applicable	Not applicable

ECS Service

Amazon Elastic Container Service

Express Mode

Clusters

Namespaces

Task definitions

Account settings

Amazon ECR

Repositories

AWS Batch

Online learning workshop

Documentation

Discover products

Subscriptions

Tell us what you think

Clusters

week5-ecs

Create service

▼ Load balancing - optional

Configure load balancing using Amazon Elastic Load Balancing to distribute traffic evenly across the healthy tasks in your service.

☒ Use load balancing

VPC

The VPC for your load balancing resources must be the same as the VPC for your service with awscli.

vpc-0a1482905e05556c

Load balancer type

info

Specify the load balancer type to distribute incoming traffic across the tasks running in your service.

☒ Application Load Balancer

As Application Load Balancer makes routing decisions at the application layer (HTTP/HTTPS), supports path-based routing, and can route requests to one or more ports.

☐ Network Load Balancer

A Network Load Balancer makes routing decisions at the transport layer (TCP/UDP).

Container

The container and port to load balance the incoming traffic to.

appstudent:8000:8000

Host port:Container port

Application Load Balancer

Specify whether to create a new load balancer or choose an existing one.

☐ Create a new load balancer

☒ Use an existing load balancer

Load balancer

Choose an existing load balancer to distribute traffic. View existing load balancers and create new one in EC2 Console.

alb-week5

alb-week5-352477752-us-east-1.elb.amazonaws.com

Internet-facing

Listener

info

Specify the port and protocol that the load balancer will listen for connection requests on.

☐ Create new listener

☒ Use an existing listener

Listener

HTTPS:443

Listener rules for 443:HTTPS (1)

Traffic received by the listener is routed according to its rules. Rules are evaluated in priority order, from the lowest value to the highest value. The default rule is evaluated last.

Domain (ACM, Route 53)

Route 53

Hosted zones

all01.ovh

Create record

Create record

info

Quick create record

Switch to wizard

Record 1

Delete

Record name

info

subdomain

all01.ovh

Record type

info

A - Routes traffic to an IPv4 address and some AWS resources

Alias

Keep blank to create a record for the root domain.

☒ Alias

Route traffic to

info

Alias to Application and Classic Load Balancer

US East (N. Virginia)

duelstack-alb-week5-352477752-us-east-1.elb.amazonaws.com

Routing policy

info

Simple routing

Evaluate target health

☒ Yes

Add another record

Cancel

Create records

► View existing records

The following table lists the existing records in all01.ovh.